API DESIGN DOCUMENT

SkillLens

Complete REST API Reference — All Endpoints, Request Bodies, and Responses

Project	SkillLens — Final Year Major Project (B.E. Computer Science)
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1. Global API Conventions

Property	Value
Base URL — development	http://localhost:5000/api
Base URL — production	https://skilllens-api.onrender.com/api
Protocol	HTTPS in production; HTTP allowed in development only
Data format	All request bodies and responses are JSON (Content-Type: application/json)
File uploads	multipart/form-data for resume upload endpoint only
Authentication	JWT Bearer token — add header: Authorization: Bearer <token>
JWT expiry	7 days from issuance
Token storage (client)	localStorage or React Context — never exposed to backend logs

Standard Success Envelope

All successful responses follow this structure:

```
{
  "success": true,
  "data": { }        // payload varies by endpoint — documented per endpoint below
}
```

Standard Error Envelope

All error responses follow this structure:

```
{
  "success": false,
  "message": "Human-readable error description",
  "code": "MACHINE_READABLE_CODE"    // optional — used for specific client handling
}
```

JWT Header — Required on All Protected Endpoints

Request header:


```
Authorization: Bearer
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiI2NjBhYmMxMjM0NTY3ODkwYWJjZGVmMTIiLCJ
lbWFpbCI6InN0dWRlbnRAZXhhbXBsZS5jb20iLCJ0YXJnZXRSb2xlIjoiYmFja2VuZCIsImlhdCI6MTcwOTgxMjg
wMCwiZXhwIjoxNzEwNDE3NjAwfQ.SIGNATURE
```


Note: Sending a request to a protected endpoint without a valid JWT returns HTTP 401 immediately — the route handler is never reached.

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2. Authentication Endpoints

Base path: /api/auth — All registration and login endpoints are public (no JWT required).

POST `/api/auth/register` Public

Create a new student account. Returns a JWT immediately on success — no separate login step required.

Request Body Fields

Field	Type	Req.	Validation	Description
`name`	String	**Yes**	Min 2 chars, max 60 chars	Full name of the student
`email`	String	**Yes**	Valid email format, unique	Email address used for login — stored in lowercase
`password`	String	**Yes**	Min 6 chars	Plain text — hashed with bcrypt (10 rounds) before storage. Never logged or returned.
`targetRole`	String	**Yes**	Enum — see allowed values	Student's target job role. Allowed: "frontend" | "backend" | "fullstack" | "data" | "java"

Request body example:

```
{
  "name": "Rahul Sharma",
  "email": "rahul.sharma@example.com",
  "password": "securepass123",
  "targetRole": "backend"
}
```

201 Created — registration successful

```
{
  "success": true,
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiI2NjBhYmMxMjM...",
    "user": {
      "_id": "660abc123456789abcdef012",
      "name": "Rahul Sharma",
      "email": "rahul.sharma@example.com",
      "targetRole": "backend",
      "createdAt": "2026-03-19T10:30:00.000Z"
    }
  }
}
```

400 Bad Request — validation failed

```
{
  "success": false,
  "message": "All fields are required.",
  "code": "MISSING_FIELDS"
}
```

409 Conflict — email already registered

```
{
  "success": false,
  "message": "An account with this email already exists.",
  "code": "EMAIL_DUPLICATE"
}
```

POST `/api/auth/login` Public

Authenticate an existing user. Returns a fresh JWT on success.

Request Body Fields

Field	Type	Req.	Validation	Description

email	String	Yes	Must match a registered account	Registered email address
password	String	Yes	Min 6 chars	Plain text password — compared against stored bcrypt hash

Request body example:

```
{
  "email": "rahul.sharma@example.com",
  "password": "securepass123"
}
```

200 OK — login successful

```
{
  "success": true,
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiI2NjBhYmMxMjM...",
    "user": {
      "_id": "660abc123456789abcdef012",
      "name": "Rahul Sharma",
      "email": "rahul.sharma@example.com",
      "targetRole": "backend",
      "createdAt": "2026-03-19T10:30:00.000Z"
    }
  }
}
```

401 Unauthorised — wrong credentials

```
{
  "success": false,
  "message": "Invalid email or password.",
  "code": "INVALID_CREDENTIALS"
}
```

GET /api/auth/profile Protected

Return the authenticated user's profile. No request body required — user is identified from the JWT.

No request body. JWT in Authorization header identifies the user.

200 OK

```
{
  "success": true,
  "data": {
    "_id": "660abc123456789abcdef012",
    "name": "Rahul Sharma",
    "email": "rahul.sharma@example.com",
    "targetRole": "backend",
    "createdAt": "2026-03-19T10:30:00.000Z",
    "updatedAt": "2026-03-19T14:20:00.000Z"
  }
}
```

PUT `/api/auth/profile` 🔒 Protected

Update the authenticated user's target job role. Triggers recalculation of company match percentages.

Request Body Fields

Field	Type	Req.	Validation	Description
`targetRole`	String	**Yes**	Enum — same as register	New target role. Allowed: "frontend" | "backend" | "fullstack" | "data" | "java"

Request body example:

```
{
  "targetRole": "fullstack"
}
```

200 OK — profile updated

```
{
  "success": true,
  "data": {
    "_id": "660abc123456789abcdef012",
    "name": "Rahul Sharma",
    "email": "rahul.sharma@example.com",
    "targetRole": "fullstack",
    "updatedAt": "2026-03-19T15:00:00.000Z"
  }
}
```

Side effects: Company match percentages recalculated for the new target role immediately after update.

3. Resume Endpoints

Base path: /api/resume — All endpoints are protected. Resume file is never persisted on disk.

POST `/api/resume/upload` Protected

Upload a PDF resume. Extracts raw text using pdf-parse, detects skills via string matching, and returns detected and missing skills. File is processed in memory and discarded — never saved to disk.

Request — multipart/form-data

Field Name	Type	Required	Constraints	Description
resume	File (PDF)	Yes	PDF only, max 5 MB, text-extractable	The student's resume file. Must be a text-based PDF — scanned image PDFs cannot be processed.

Note: Content-Type for this endpoint must be multipart/form-data — NOT application/json.

200 OK — skills extracted successfully

```
{
  "success": true,
  "data": {
    "detectedSkills": [
      "react",
      "nodejs",
      "mongodb",
      "git",
      "restapi"
    ],
    "missingSkills": [
      "dsa",
      "sql",
      "systemdesign"
    ],
    "roleRequiredSkills": [
      "nodejs",
      "mongodb",
      "dsa",
      "sql",
      "restapi",
      "systemdesign",
      "git"
    ],
    "coveragePercent": 57,
    "message": "5 of 7 required skills detected in your resume for the backend role."
  }
}
```

400 Bad Request — no file or wrong format

```
{
  "success": false,
  "message": "Please upload a PDF file.",
  "code": "INVALID_FILE_TYPE"
}
```

422 Unprocessable Entity — text extraction failed

```
{
  "success": false,
  "message": "We could not extract text from this PDF. Please ensure you are uploading
a text-based PDF, not a scanned image.",
  "code": "PDF_EXTRACTION_FAILED"
}
```

Side effects: No side effects — resume data is not stored. User is prompted to take skill quizzes for detected skills on the frontend.

---

4. Quiz Engine Endpoints

Base path: /api/quiz — All endpoints are protected.

GET /api/quiz/scores Protected

Return all skill depth scores for the authenticated user. Returns an empty array if no quizzes have been taken yet.

No request body required.

200 OK

```
{
  "success": true,
  "data": {
    "scores": [
      {
        "skill": "react",
        "depthScore": 70,
        "depthLevel": "Intermediate",
        "quizAttempts": 2,
        "lastAttempt": "2026-03-18T09:15:00.000Z"
      },
      {
        "skill": "nodejs",
        "depthScore": 50,
        "depthLevel": "Intermediate",
        "quizAttempts": 1,
        "lastAttempt": "2026-03-17T14:00:00.000Z"
      },
      {
        "skill": "mongodb",
        "depthScore": 30,
        "depthLevel": "Beginner",
        "quizAttempts": 1,
        "lastAttempt": "2026-03-16T11:00:00.000Z"
      }
    ],
    "totalSkillsTaken": 3,
    "averageDepthScore": 50
  }
}
```

GET /api/quiz/:skill Protected

Fetch 10 randomised MCQ questions for a specific skill. Correct answers are stripped before sending — they are only used server-side during answer submission.

URL Parameters

Parameter	Type	Allowed Values	Description
`skill`	String	"react" | "nodejs" | "mongodb" | "sql" | "python" | "java" | "dsa" | "restapi" | "systemdesign" | "git"	The skill identifier to load questions for

No request body required. Skill passed as URL parameter.

200 OK — 10 randomised questions returned

```
{
  "success": true,
  "data": {
    "skill": "nodejs",
    "totalQuestions": 10,
    "questions": [
      {
        "id": 7,
        "question": "What does the Express.js middleware function next() do?",
        "options": [
          "A. Terminates the current request",
          "B. Passes control to the next middleware function in the stack",
          "C. Sends a response to the client",
          "D. Restarts the Express server"
        ]
      },
      {
        "id": 3,
        "question": "Which method is used to parse incoming JSON request bodies in
Express.js?",
        "options": [
          "A. app.use(express.text())",
          "B. app.use(express.json())",
          "C. app.use(express.urlencoded())",
          "D. app.use(express.raw())"
        ]
      }
    ]
  }
}
```

404 Not Found — invalid skill name

```
{
  "success": false,
  "message": "Skill \"invalidskill\" not found. Valid skills are: react, nodejs,
mongodb, sql, python, java, dsa, restapi, systemdesign, git.",
  "code": "SKILL_NOT_FOUND"
}
```

POST `/api/quiz/:skill/submit` Protected

Submit answers for a completed quiz. Server loads correct answers, scores the submission, classifies depth level, and stores or updates the SkillProfile for this user.

URL Parameters

Parameter	Type	Allowed Values	Description
`skill`	String	Same as GET /quiz/:skill	The skill being submitted

Request Body Fields

Field	Type	Req.	Validation	Description
`answers`	Object	**Yes**	Exactly 10 key-value pairs	Keys are question IDs (as strings or numbers). Values are the selected option index (0–3) or the selected option text. Must include an answer for every question returned.

Request body example:

```
{
  "answers": {
    "7":  1,
    "3":  1,
    "12": 0,
    "5":  2,
    "9":  3,
    "1":  1,
    "15": 0,
    "8":  2,
    "4":  1,
    "11": 3
  }
}
```

200 OK — quiz scored and stored

```
{
  "success": true,
  "data": {
    "skill": "nodejs",
    "correct": 7,
    "wrong": 3,
    "total": 10,
    "depthScore": 70,
    "depthLevel": "Intermediate",
    "previousScore": 50,
    "previousLevel": "Intermediate",
    "improved": true,
    "scoreDelta": 20,
    "jobMinimumForRole": 65,
    "meetsJobMinimum": true,
    "message": "Good improvement! You now meet the minimum depth required for your
backend role target."
  }
}
```

400 Bad Request — wrong number of answers

```
{
  "success": false,
  "message": "Expected 10 answers, received 8. Please answer all questions before
submitting.",
  "code": "INCOMPLETE_SUBMISSION"
}
```

Side effects: 1) SkillProfile document created or updated for this user+skill. 2) Company match percentages recalculated for all 25 companies. 3) Readiness Score recalculated and new history entry saved.

5. Company Match Endpoints

Base path: /api/companies — All endpoints are protected. Match percentages are computed on-demand from stored SkillProfile data — not cached.

GET `/api/companies` Protected

Return all 25 company profiles with the authenticated user's current match percentage for each. Sorted by matchPercent descending.

No request body or parameters required.

200 OK — full company list with match percentages

```
{
  "success": true,
  "data": {
    "companies": [
      {
        "id": "tcs",
        "name": "TCS",
        "sector": "Service",
        "type": "Indian MNC",
        "matchPercent": 78,
        "readinessLabel": "Good match",
        "requiredSkillCount": 5,
        "studentMetCount": 4
      },
      {
        "id": "wipro",
        "name": "Wipro",
        "sector": "Service",
        "type": "Indian MNC",
        "matchPercent": 71,
        "readinessLabel": "Good match",
        "requiredSkillCount": 5,
        "studentMetCount": 3
      },
      {
        "id": "razorpay",
        "name": "Razorpay",
        "sector": "Fintech",
        "type": "Indian Product",
        "matchPercent": 31,
        "readinessLabel": "Needs work",
        "requiredSkillCount": 6,
        "studentMetCount": 2
      }
    ],
    "totalCompanies": 25,
    "strongMatches": 8,
    "bestMatch": {
      "id": "tcs",
      "name": "TCS",
      "matchPercent": 78
    }
  }
}
```

GET `/api/companies/:id` Protected

Return the full profile of one company, overlaid with the student's current depth scores for each required skill.

URL Parameters

Parameter	Type	Allowed Values	Description
`id`	String	"tcs" | "infosys" | "wipro" | "hcl" | "cognizant" | "techmahindra" | "capgemini" | "accenture" | "hexaware" | "mphasis" | "flipkart" | "zomato" | "swiggy" | "razorpay" | "phonepe" | "meesho" | "paytm" | "cred" | "juspay" | "freshworks" | "browserstack" | "postman" | "zoho" | "persistent" | "mindtree"	Company identifier string from companies.json

No request body required.

200 OK — full company profile with student depth overlay


```
{
  "success": true,
  "data": {
    "id": "razorpay",
    "name": "Razorpay",
    "sector": "Fintech",
    "type": "Indian Product",
    "matchPercent": 31,
    "requiredSkills": [
      {
        "skill": "nodejs",
        "weight": 25,
        "minimumDepth": 75,
        "studentDepth": 50,
        "depthLevel": "Intermediate",
        "met": false,
        "gap": 25
      },
      {
        "skill": "dsa",
        "weight": 30,
        "minimumDepth": 70,
        "studentDepth": 38,
        "depthLevel": "Beginner",
        "met": false,
        "gap": 32
      },
      {
        "skill": "mongodb",
        "weight": 20,
        "minimumDepth": 60,
        "studentDepth": 0,
        "depthLevel": "Not taken",
        "met": false,
        "gap": 60
      },
      {
        "skill": "restapi",
        "weight": 15,
        "minimumDepth": 60,
        "studentDepth": 70,
        "depthLevel": "Intermediate",
        "met": true,
        "gap": 0
      },
      {
```

```
        "skill": "systemdesign",
        "weight": 10,
        "minimumDepth": 50,
        "studentDepth": 0,
        "depthLevel": "Not taken",
        "met": false,
        "gap": 50
      }
    ],
    "verdict": "You meet 1 of 5 required skills for Razorpay. Focus on DSA first — it
has the highest weight and your biggest gap."
  }
}
```

404 Not Found — invalid company ID

```
{
  "success": false,
  "message": "Company \"invalidco\" not found.",
  "code": "COMPANY_NOT_FOUND"
}
```

GET `/api/companies/:id/gap` Protected

Return a prioritised gap analysis for one company — the top 3 skills dragging the match score down, the impact of fixing each, and a plain-language next-step verdict.

URL Parameters

Parameter	Type	Allowed Values	Description
`id`	String	Same as GET /companies/:id	Company identifier string

No request body required.

```

```

200 OK — gap analysis

```
{
  "success": true,
  "data": {
    "company": "Razorpay",
    "companyId": "razorpay",
    "currentMatchPercent": 31,
    "topGaps": [
      {
        "rank": 1,
        "skill": "dsa",
        "skillWeight": 30,
        "studentDepth": 38,
        "minimumDepth": 70,
        "depthGap": 32,
        "scoreImpact": 18,
        "reason": "Your DSA depth is 38% — Razorpay requires 70% minimum.",
        "fixedMatchPercent": 49,
        "actionToFix": "Retake the DSA quiz after completing the DSA roadmap tasks."
      },
      {
        "rank": 2,
        "skill": "nodejs",
        "skillWeight": 25,
        "studentDepth": 50,
        "minimumDepth": 75,
```

```
        "depthGap": 25,
        "scoreImpact": 8,
        "reason": "Your Node.js depth is 50% — Razorpay requires 75% minimum.",
        "fixedMatchPercent": 57,
        "actionToFix": "Retake the Node.js quiz after studying middleware, async
patterns, and error handling."
      },
      {
        "rank": 3,
        "skill": "mongodb",
        "skillWeight": 20,
        "studentDepth": 0,
        "minimumDepth": 60,
        "depthGap": 60,
        "scoreImpact": 14,
        "reason": "MongoDB quiz not yet taken — skill shows as missing.",
        "fixedMatchPercent": 71,
        "actionToFix": "Take the MongoDB quiz. Even reaching Intermediate level (60%)
would add 14% to your match score."
      }
    ],
    "projectedMatchIfAllFixed": 87,
    "verdict": "Fix DSA depth first — it is your highest-weight gap. Then take the
MongoDB quiz — even beginner level will move your score significantly."
  }
}
```

6. Interview Endpoints

Base path: /api/interview — All endpoints are protected. AI evaluation uses OpenAI gpt-3.5-turbo.

GET `/api/interview/questions/:role` Protected

Load 5 role-specific interview questions for a mock session.

URL Parameters

Parameter	Type	Allowed Values	Description
`role`	String	"frontend" | "backend" | "fullstack" | "data" | "java"	The target role to load questions for. Questions are role-specific and cover core technical topics.

No request body required.

200 OK — 5 questions returned


```
{
  "success": true,
  "data": {
    "role": "backend",
    "totalQuestions": 5,
    "sessionId": "sess_660abc123456",
    "questions": [
      {
        "id": 1,
        "question": "Explain the difference between authentication and authorisation.
Give a real-world example.",
        "topic": "security",
        "difficulty": "medium"
      },
      {
        "id": 2,
        "question": "What is middleware in Express.js? Write a simple example of a
custom middleware function.",
        "topic": "nodejs",
        "difficulty": "medium"
      },
      {
        "id": 3,
        "question": "What is the difference between SQL and NoSQL databases? When
would you choose MongoDB over MySQL?",
        "topic": "database",
        "difficulty": "easy"
      },
      {
        "id": 4,
        "question": "Explain what a REST API is and list the four main HTTP methods
with their purpose.",
        "topic": "restapi",
        "difficulty": "easy"
      },
      {
        "id": 5,
        "question": "What is an event loop in Node.js? How does it handle asynchronous
operations?",
        "topic": "nodejs",
        "difficulty": "hard"
      }
    ]
  }
}
```

POST /api/interview/evaluate Protected

Submit a single question-answer pair for AI evaluation. Calls OpenAI with a structured prompt that enforces a JSON response. Stores the result and feeds missed keywords back to the skill profile system.

Request Body Fields

Field	Type	Req.	Validation	Description
sessionId	String	Yes	From GET /interview/questions/:role	Session identifier returned when questions were loaded
questionId	Number	Yes	Integer 1–5	ID of the question being answered
question	String	Yes	Non-empty string	Full question text — used in the AI prompt for context
answer	String	Yes	Min 10 chars, max 3000 chars	The student's written answer to be evaluated
role	String	Yes	Enum — same as GET questions	Target role context sent to AI for evaluation framing
topic	String	No	From question object	Topic tag of the question — used for keyword-to-skill mapping

Request body example:

```
{
  "sessionId": "sess_660abc123456",
  "questionId": 1,
  "question": "Explain the difference between authentication and authorisation. Give a real-world example.",
  "answer": "Authentication is about verifying who you are. For example, when you log in with your email and password, the system checks if you are who you claim to be. Authorisation is about what you are allowed to do. After logging in, you might only have access to your own profile, not other users data. JWT is used for authentication in Node.js apps.",
  "role": "backend",
  "topic": "security"
}
```

200 OK — AI evaluation complete

```
{
  "success": true,
  "data": {
    "questionId": 1,
    "question": "Explain the difference between authentication and authorisation. Give a real-world example.",
    "answer": "Authentication is about verifying who you are...",
    "evaluation": {
      "technicalScore": 7,
      "communicationScore": 6,
      "technicalFeedback": "Correctly distinguishes authentication from authorisation and provides a valid real-world example. However, the answer missed token expiry, refresh tokens, and role-based access control — all important in a backend context.",
      "communicationFeedback": "Answer is clear and structured with a good use of an example. Could be improved by stating the core distinction upfront in one sentence before expanding.",
      "missedPoint": "Role-Based Access Control (RBAC) and JWT token expiry / refresh strategy",
      "improvedAnswer": "Authentication verifies identity — confirming you are who you claim to be, typically via credentials. Authorisation determines permissions — what resources an authenticated user can access, often enforced via roles (RBAC) and implemented using middleware that validates JWT claims on each request.",
      "keywordsMissed": [
```

```
        "RBAC",
        "token expiry",
        "refresh token",
        "role-based",
        "middleware"
      ]
    },
    "storedResultId": "660xyz987654321fedcba098",
    "averageScore": 6.5
  }
}
```

503 Service Unavailable — OpenAI API unreachable

```
{
  "success": false,
  "message": "AI evaluation is temporarily unavailable. Your answer has been saved and
will not be scored for this session.",
  "code": "AI_SERVICE_UNAVAILABLE",
  "data": {
    "answerSaved": true,
    "storedResultId": "660xyz987654321fedcba098",
    "scored": false
  }
}
```

Side effects: 1) InterviewResult document saved with all evaluation fields. 2) keywordsMissed array processed — each keyword mapped to its parent skill, SkillProfile weakFlag updated. 3) Readiness Score recalculated after all 5 questions submitted.

GET `/api/interview/history` Protected

Return all past interview results for the authenticated user, sorted newest first.

No request body or parameters required.

```

```

200 OK

```
{
  "success": true,
  "data": {
    "totalSessions": 3,
    "averageTechnicalScore": 6.8,
    "averageCommunicationScore": 5.4,
    "results": [
      {
        "_id": "660xyz987654321fedcba098",
        "role": "backend",
        "question": "Explain the difference between authentication and
authorisation.",
        "topic": "security",
        "technicalScore": 7,
        "communicationScore": 6,
        "technicalFeedback": "Correctly distinguishes authentication from
authorisation...",
        "communicationFeedback": "Clear structure, good example used.",
        "missedPoint": "RBAC and JWT token expiry / refresh strategy",
        "improvedAnswer": "Authentication verifies identity...",
        "keywordsMissed": ["RBAC", "token expiry", "refresh token"],
        "date": "2026-03-19T11:00:00.000Z"
      }
    ]
  }
}
```

7. Roadmap Endpoints

Base path: /api/roadmap — All endpoints are protected. AI is used only to order skills; resource links come from the static resources.json library.

POST `/api/roadmap/generate` Protected

Generate a personalised week-by-week roadmap. Calls OpenAI to determine skill priority order based on gap analysis, then populates each week with curated resources from resources.json. Previous roadmap for this user is replaced.

Request Body Fields

Field	Type	Req.	Validation	Description
`targetCompany`	String	**Yes**	Valid company ID from companies.json	The company to optimise the roadmap for. Gap analysis for this company drives the skill ordering.
`weeksLeft`	Number	**Yes**	Integer 2–20	Number of weeks available before placement season. Determines how many weeks the roadmap spans.

Request body example:

```
{
  "targetCompany": "razorpay",
  "weeksLeft": 8
}
```

200 OK — roadmap generated and stored

```
{
  "success": true,
  "data": {
    "targetCompany": "razorpay",
    "targetCompanyName": "Razorpay",
    "totalWeeks": 8,
    "generatedAt": "2026-03-19T12:00:00.000Z",
    "weeks": [
      {
        "week": 1,
        "skill": "dsa",
        "level": "beginner",
        "reason": "DSA has the highest weight (30%) at Razorpay and your current depth
is only 38% — it is your biggest score gap.",
        "resources": {
          "video": {
            "title": "DSA Full Course for Beginners — Striver",
            "url":
"https://www.youtube.com/playlist?list=PLgUwDviBIf0oF6QL8m22w1hIDC1vJ_BHz",
            "duration": "~6 hours"
          },
          "article": {
            "title": "Data Structures Tutorial — GeeksforGeeks",
            "url": "https://www.geeksforgeeks.org/data-structures/"
          },
          "practice": {
            "title": "LeetCode — Easy Array problems",
            "url": "https://leetcode.com/tag/array/",
            "targetProblems": 10
          },
          "task": {
            "description": "Implement a stack and queue from scratch without using
built-in methods. Write both in Node.js and test with at least 5 operations each.",
            "estimatedHours": 3
          }
        },
```

```
        "completed": false,
        "completedAt": null
      },
      {
        "week": 2,
        "skill": "dsa",
        "level": "intermediate",
        "reason": "Continue DSA at intermediate level to reach the 70% minimum
Razorpay requires.",
        "resources": {
          "video": {
            "title": "Binary Search, Recursion, Linked Lists — Striver",
            "url":
"https://www.youtube.com/playlist?list=PLgUwDviBIf0oF6QL8m22w1hIDC1vJ_BHz",
            "duration": "~5 hours"
          },
          "article": {
            "title": "Top 50 Problems on Linked List — GFG",
            "url": "https://www.geeksforgeeks.org/top-50-linked-list-problems/"
          },
          "practice": {
            "title": "LeetCode — Medium Array and Linked List",
            "url": "https://leetcode.com/tag/linked-list/",
            "targetProblems": 8
          },
          "task": {
            "description": "Solve 8 Medium-level LeetCode problems covering arrays,
strings, and linked lists. Record time taken per problem.",
            "estimatedHours": 4
          }
        },
        "completed": false,
        "completedAt": null
      }
    ]
  }
}
```


Side effects: Previous roadmap for this user deleted and replaced. New RoadmapProgress documents created — one per week. Readiness Score roadmapCompletion component reset to 0.

GET `/api/roadmap` Protected

Return the current roadmap for the authenticated user with live completion status.

No request body or parameters required.

200 OK — current roadmap returned


```
{
  "success": true,
  "data": {
    "targetCompany": "razorpay",
    "targetCompanyName": "Razorpay",
    "totalWeeks": 8,
    "completedWeeks": 2,
    "completionPercent": 25,
    "generatedAt": "2026-03-10T12:00:00.000Z",
    "weeks": [
      {
        "week": 1,
        "skill": "dsa",
        "level": "beginner",
        "reason": "Highest weight gap for Razorpay.",
        "resources": { },
```

```
        "completed": true,
        "completedAt": "2026-03-14T18:00:00.000Z"
      },
      {
        "week": 2,
        "skill": "dsa",
        "level": "intermediate",
        "reason": "Continue DSA to reach minimum depth.",
        "resources": { },
        "completed": true,
        "completedAt": "2026-03-19T10:00:00.000Z"
      },
      {
        "week": 3,
        "skill": "mongodb",
        "level": "beginner",
        "reason": "MongoDB quiz not taken — adds 14% to Razorpay score at beginner
level.",
        "resources": { },
        "completed": false,
        "completedAt": null
      }
    ]
  }
}
```


404 Not Found — no roadmap generated yet


```
{
  "success": false,
  "message": "No roadmap found. Please generate a roadmap first using POST
/api/roadmap/generate.",
  "code": "ROADMAP_NOT_FOUND"
}
```

PUT `/api/roadmap/:week/complete` Protected

Mark a specific week as complete. Self-reported by the student. Immediately triggers Readiness Score recalculation.

URL Parameters

Parameter	Type	Allowed Values	Description
`week`	Number	Integer — 1 to totalWeeks	The week number to mark as complete

No request body required. Week number passed as URL parameter.

200 OK — week marked complete


```
{
  "success": true,
  "data": {
    "week": 3,
    "skill": "mongodb",
    "level": "beginner",
    "completed": true,
    "completedAt": "2026-03-19T16:30:00.000Z",
    "previousCompletionPercent": 25,
    "newCompletionPercent": 37,
    "previousReadinessScore": 58,
```

```
    "newReadinessScore": 61,
    "message": "Week 3 complete! Your Readiness Score increased by 3 points."
  }
}
```

404 Not Found — week number invalid

```
{
  "success": false,
  "message": "Week 9 not found in your current roadmap. Your roadmap has 8 weeks.",
  "code": "WEEK_NOT_FOUND"
}
```

Side effects: RoadmapProgress document updated (completed=true, completedAt=now). Readiness Score recalculated immediately.

8. Dashboard Endpoint

Base path: /api/dashboard — Protected. Returns all data needed to render the full dashboard in a single API call — avoids multiple parallel calls from the frontend.

GET `/api/dashboard/metrics` Protected

Aggregate and return all dashboard data: Readiness Score and history, skill depths, company matches, interview averages, roadmap status, and the next recommended action.

No request body or parameters required. User identified from JWT.

200 OK — full dashboard payload


```
{
  "success": true,
  "data": {
    "readiness": {
      "score": 62,
      "label": "Almost there",
      "components": {
        "avgDepthScore": 56,
        "avgInterviewScore": 65,
        "roadmapCompletion": 37,
        "bestCompanyMatch": 78
      },
      "weights": {
        "avgDepthScore": 0.35,
        "avgInterviewScore": 0.30,
        "roadmapCompletion": 0.20,
        "bestCompanyMatch": 0.15
      },
      "history": [
        { "date": "2026-03-01T00:00:00.000Z", "score": 24 },
        { "date": "2026-03-08T00:00:00.000Z", "score": 38 },
        { "date": "2026-03-15T00:00:00.000Z", "score": 52 },
        { "date": "2026-03-19T00:00:00.000Z", "score": 62 }
      ]
    },
    "skills": {
      "totalTaken": 5,
      "averageDepthScore": 56,
      "scores": [
        { "skill": "react",        "depthScore": 70, "depthLevel": "Intermediate",
"lastAttempt": "2026-03-18T09:15:00.000Z" },
        { "skill": "nodejs",       "depthScore": 70, "depthLevel": "Intermediate",
"lastAttempt": "2026-03-17T14:00:00.000Z" },
        { "skill": "mongodb",      "depthScore": 30, "depthLevel": "Beginner",
"lastAttempt": "2026-03-16T11:00:00.000Z" },
        { "skill": "dsa",          "depthScore": 60, "depthLevel": "Intermediate",
"lastAttempt": "2026-03-19T10:00:00.000Z" },
        { "skill": "restapi",      "depthScore": 50, "depthLevel": "Intermediate",
"lastAttempt": "2026-03-15T08:00:00.000Z" }
      ]
    },
    "companies": {
      "topMatches": [
        { "id": "tcs",      "name": "TCS",      "matchPercent": 78, "sector":
"Service" },
        { "id": "wipro",    "name": "Wipro",    "matchPercent": 71, "sector":
"Service" },
        { "id": "infosys",  "name": "Infosys",  "matchPercent": 68, "sector":
"Service" }
      ],
      "bestMatch": { "id": "tcs", "name": "TCS", "matchPercent": 78 }
    },
    "interviews": {
```

```
      "totalAnswered": 12,
      "averageTechnicalScore": 6.8,
      "averageCommunicationScore": 5.4,
      "averageCombinedScore": 6.1,
      "recentSession": {
        "role": "backend",
        "date": "2026-03-19T11:00:00.000Z",
        "avgScore": 6.5
      }
    },
    "roadmap": {
      "active": true,
      "targetCompany": "razorpay",
      "targetCompanyName": "Razorpay",
      "totalWeeks": 8,
      "completedWeeks": 3,
      "completionPercent": 37,
      "currentWeek": {
        "week": 4,
        "skill": "nodejs",
        "level": "intermediate",
        "task": "Build a REST API with JWT authentication — register, login, and one
protected route."
      }
    },
    "nextStep": {
      "action": "Retake the MongoDB quiz",
      "reason": "MongoDB is your weakest skill (30% depth) and has a 20% weight at
Razorpay — it is the single highest-impact improvement available.",
      "expectedScoreGain": 7
    }
  }
}
```


9. HTTP Status Code Reference

Code	Status	When Returned in SkillLens	Response body key
200	OK	Successful GET and PUT requests	success: true, data: { }
201	Created	POST /api/auth/register — new account created	success: true, data: { token, user }
400	Bad Request	Missing required fields, invalid file type, wrong answer count, malformed JSON	success: false, message, code
401	Unauthorised	No JWT provided, expired JWT, or tampered JWT on any protected route	success: false, message, code: AUTH_REQUIRED
403	Forbidden	Valid JWT but user trying to access another user's data	success: false, message, code: FORBIDDEN
404	Not Found	Company ID not found, quiz skill not found, roadmap not generated yet	success: false, message, code
409	Conflict	POST /api/auth/register with an already-registered email	success: false, message, code: EMAIL_DUPLICATE
422	Unprocessable Entity	PDF uploaded successfully but text extraction failed — likely a scanned image PDF	success: false, message, code: PDF_EXTRACTION_FAILED
500	Internal Server Error	Unexpected server error — database unreachable, unhandled exception	success: false, message, code: INTERNAL_ERROR
503	Service Unavailable	OpenAI API unreachable during POST /api/interview/evaluate	success: false, message, code: AI_SERVICE_UNAVAILABLE, data: { answerSaved: true }

10. Complete Endpoint Index

Method	Endpoint	Auth	Summary
POST	/api/auth/register	No	Register new user account — returns JWT
POST	/api/auth/login	No	Login — returns JWT
GET	/api/auth/profile	Yes	Get authenticated user profile
PUT	/api/auth/profile	Yes	Update target role
POST	/api/resume/upload	Yes	Upload PDF — extract and return detected skills
GET	/api/quiz/scores	Yes	Get all skill depth scores for user
GET	/api/quiz/:skill	Yes	Fetch 10 randomised MCQs for a skill
POST	/api/quiz/:skill/submit	Yes	Submit quiz answers — score and store depth
GET	/api/companies	Yes	All 25 companies with match percentages
GET	/api/companies/:id	Yes	Full company profile with student depth overlay
GET	/api/companies/:id/gap	Yes	Gap analysis — top 3 skill gaps for a company
GET	/api/interview/questions/:role	Yes	Load 5 interview questions for a role
POST	/api/interview/evaluate	Yes	Submit answer — AI evaluation and storage
GET	/api/interview/history	Yes	All past interview results for user
POST	/api/roadmap/generate	Yes	Generate personalised roadmap
GET	/api/roadmap	Yes	Get current roadmap with completion status
PUT	/api/roadmap/:week/complete	Yes	Mark a roadmap week as complete
GET	/api/dashboard/metrics	Yes	Full dashboard payload — all modules aggregated